TEXORA · LECTURE SHEET

Introduction to Engineering Materials

The families of materials and why they behave as they do: classification, atomic bonding, crystal structures and unit cells, packing factors and theoretical density, and the structure–property idea that runs through this whole course.

ME-1207 · ENGINEERING MATERIALS · TOPIC 1 · LEVEL 1 TERM 2

1. The families

Materials science begins with a claim: **properties are decided by structure** — the arrangement of atoms and the bonds between them. Engineering materials fall into families whose members share bonding and therefore behaviour:

FAMILY	BONDING / STRUCTURE	SIGNATURE PROPERTIES	EXAMPLES
Metals	Metallic bond; crystalline	Strong, ductile, conduct heat and electricity, opaque	Steel, aluminium, copper
Ceramics	Ionic/covalent; crystalline or glassy	Very hard, stiff, heat- and corrosion-resistant, brittle, insulating	Alumina, glass, porcelain, cement
Polymers	Covalent chains + weak secondary bonds between them	Light, flexible, insulating, low melting/softening	Polyethylene, nylon, polyester (PET)
Composites	Two families combined deliberately	Designed properties — light *and* strong	GFRP, carbon-fibre epoxy, reinforced concrete
Semiconductors	Covalent with a small band gap	Conductivity tunable by doping and light	Silicon, germanium

The textile engineer lives among polymers (every synthetic fibre is one; cotton and wool are natural polymers) but builds the machines from metals, guides yarn across ceramic eyelets, and increasingly designs textile composites — this course's later topics tour each family in turn.

2. Bonding in one paragraph each

- **Metallic:** valence electrons are pooled into a shared “sea”; non-directional, so atoms can slide past each other (ductility) and electrons move freely (conductivity).
- **Ionic:** electrons transferred, ions packed by electrostatic attraction; strong but directional in effect — slip brings like charges together, so ionic solids cleave rather than bend (brittleness).
- **Covalent:** electrons shared in directed bonds; very strong within molecules/networks (diamond is the extreme).
- **Secondary (van der Waals, hydrogen bonds):** weak attractions between molecules or chain segments — they set polymers’ softening points and water uptake, and they hold cellulose chains together in cotton.

3. Crystal structures of metals

Metals solidify into regular lattices. Three unit cells dominate:

STRUCTURE	ATOMS PER CELL	COORDINATION NO.	APF	METALS
BCC (body-centred cubic)	2	8	0.68	α-iron (room temperature), Cr, W, V
FCC (face-centred cubic)	4	12	0.74	Al, Cu, Ni, γ-iron, austenitic stainless steel
HCP (hexagonal close-packed)	6	12	0.74	Mg, Zn, Ti (α), Co

The **atomic packing factor** APF = (volume of atoms in the cell)/(cell volume) measures how full the structure is. For BCC, atoms touch along the body diagonal: $a = 4R/\sqrt{3}$, and $APF = 2 \times (4/3)\pi R^3/a^3 = 0.68$. FCC touches along the face diagonal, $a = 4R/\sqrt{2}$, giving the close-packed limit 0.74. FCC metals are usually the most ductile — their close-packed planes offer many easy slip systems, which is why copper and aluminium form so readily while HCP magnesium, with few slip systems, is stubborn at room temperature. **Allotropy** lets one metal change camps: iron is BCC below 912 °C, FCC between 912 °C and 1,394 °C — the switch on which all steel heat treatment (ME-1208) is built.

WORKED EXAMPLE — THEORETICAL DENSITY OF IRON

$\rho = nA/(V_C N_A)$, with $n = 2$ (BCC), $A = 55.85$ g mol^{-1}, $a = 2.866$ Å $= 2.866 \times 10^{-8}$ cm, $N_A = 6.022 \times 10^{23}$.

$V_C = a^3 = 2.354 \times 10^{-23}$ cm^3. $\rho = (2 \times 55.85)/(2.354 \times 10^{-23} \times 6.022 \times 10^{23}) = 111.7/14.18 =$ **7.88**

g cm⁻³ — measured 7.87. The same recipe gives copper (FCC, a = 3.615 Å) as 8.93 g cm⁻³ against 8.96 measured: crystal geometry predicts density to a percent.

Non-crystalline solids (glasses, most polymers as used, rubber) lack long-range order; they soften over a range instead of melting sharply, and polymers add their own architecture — chain length, branching, cross-linking — on top of atomic structure. Defects matter as much as the ideal lattice: vacancies, interstitials, dislocations and grain boundaries are where real properties (strength, diffusion, corrosion attack) are actually decided; Topic 2 picks this up through mechanical behaviour.

4. Choosing a material

Selection weighs property against duty and cost: strength and stiffness for structure, density where weight counts (aerospace pays for titanium, 4.43 g cm⁻³, over steel, 7.85), temperature and corrosion limits, manufacturability, availability, recyclability. The modern method (Ashby charts) plots two properties at a time — strength against density, say — and reads off the family that fits the duty, which is why aircraft are aluminium and carbon-composite, pressure vessels are steel, and yarn guides are ceramic: each sits in a different corner of the chart.

5. Key points

- Families: metals, ceramics, polymers, composites, semiconductors — bonding predicts behaviour.
- BCC/FCC/HCP: atoms per cell 2/4/6, coordination 8/12/12, APF 0.68/0.74/0.74.
- ρ = nA/(V_C N_A) computes density from the unit cell — verify iron 7.88, copper 8.93 g cm⁻³.
- FCC's many slip systems → ductility; HCP's few → poor cold formability; iron's BCC ↔ FCC switch enables steel heat treatment.
- Real properties live in defects and microstructure, not only in the ideal lattice.

6. Practice

Answer in your own words; brief model answers follow.

1. Show that the atomic packing factor of a simple cubic cell is 0.52. (Atoms touch along the cell edge; one atom per cell.)
2. Copper is FCC with a = 3.615 Å and atomic mass 63.55. Compute its theoretical density.
3. Copper can be drawn into fine wire at room temperature, but magnesium (HCP) cracks under the same treatment. Explain through crystal structure.
4. Classify each as metal, ceramic, polymer or composite: window glass, nylon-6 filament, carbon-fibre/epoxy bicycle frame, alumina yarn guide, brass tap, Portland cement.

5. Why do polymers soften at far lower temperatures than ceramics, although both involve covalent bonding?
6. Steel is chosen over aluminium for a loom's main frame, but aluminium for its moving heddle frames. Justify both choices by properties.

Answer key

1. Edge $a = 2R$. APF $= (4/3)\pi R^3/(2R)^3 = (4/3)\pi R^3/8R^3 = \pi/6 = 0.524 \approx 0.52$. (No common metal adopts so wasteful a structure — polonium is the textbook exception.)
2. $n = 4$. $V_C = (3.615 \times 10^{-8})^3 = 4.724 \times 10^{-23}$ cm³. $\rho = 4 \times 63.55/(4.724 \times 10^{-23} \times 6.022 \times 10^{23}) = 254.2/28.45 = 8.93$ g cm^{-3}.
3. Ductility needs slip on many planes. FCC copper has close-packed {111} planes in four orientations — twelve slip systems — so dislocations always find an easy path. HCP magnesium slips readily only on its basal plane at room temperature (few independent systems), so imposed shapes exhaust slip quickly and the metal cracks instead of flowing.
4. Window glass — ceramic (glassy). Nylon-6 — polymer. Bicycle frame — composite. Alumina guide — ceramic. Brass tap — metal (alloy). Portland cement — ceramic.
5. In ceramics the covalent/ionic bonding runs continuously through the whole solid — softening means breaking primary bonds. In a polymer the strong covalent bonds run only along each chain; between chains there are only weak secondary bonds, and heat (or a solvent) can slide chains past each other at modest temperatures. The melting behaviour is set by the weakest link in the structure.
6. The main frame wants stiffness, strength, vibration damping and low cost at a place where weight is irrelevant — steel wins on all counts, and its mass even steadies the machine. Heddle frames reverse thousands of times per minute; inertia loads scale with mass, so aluminium's one-third density cuts dynamic forces, energy use and wear while still stiff and strong enough for the lighter duty.
